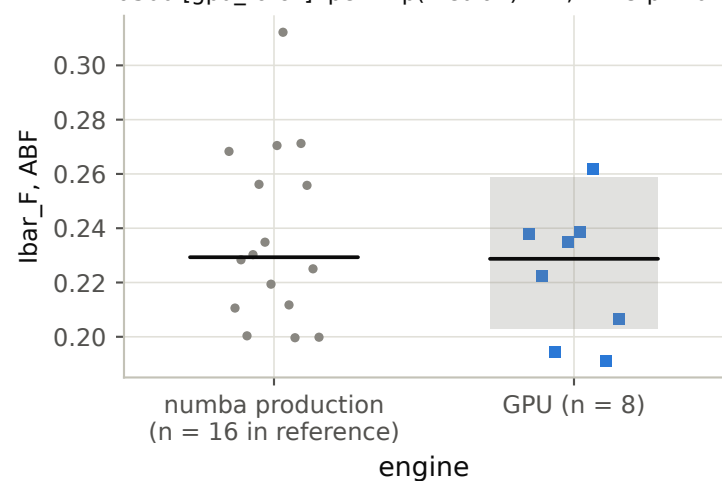


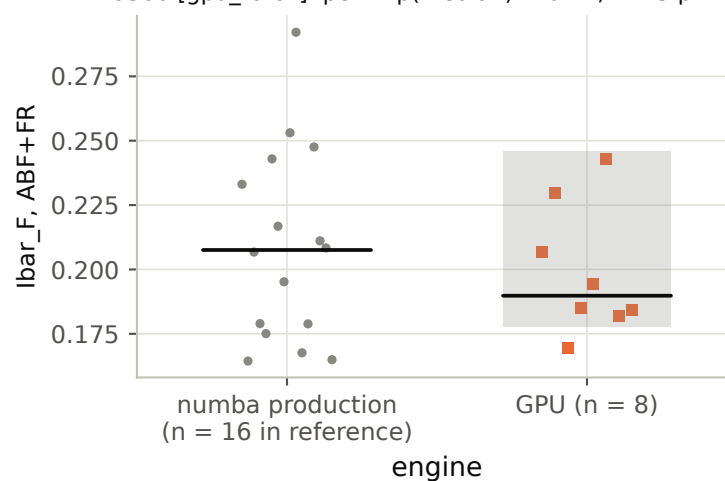
EXPLORATORY (tolerance stated in analyze_parallel_benchmark.py, not in the plan): torch backend vs validated numba production, N = 512

verdict (Holm over R1-R5, ≥ 8 pairs): CONSISTENT

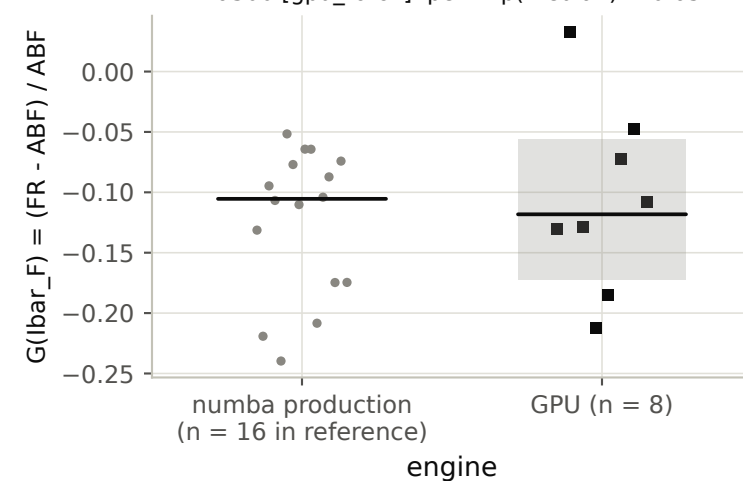
lta300 [gpu_torch]: perm. $p(\text{median}) = 1$, MWU $p = 0.45$



lta300 [gpu_torch]: perm. $p(\text{median}) = 0.47$, MWU $p = 0.74$

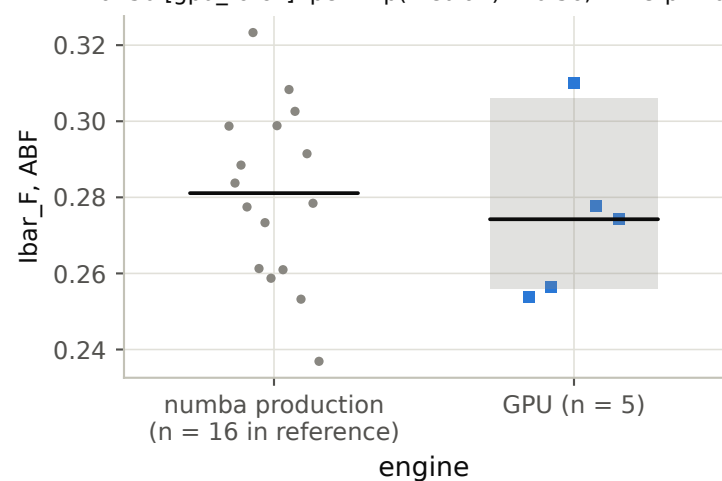


lta300 [gpu_torch]: perm. $p(\text{median}) = 0.65$

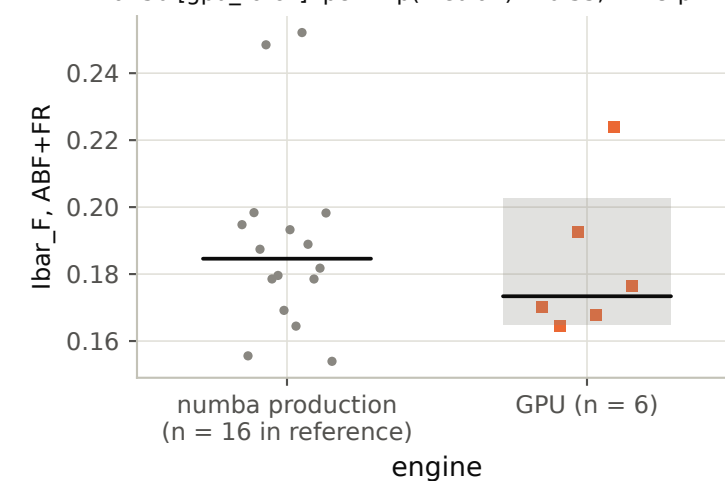


verdict (Holm over R1-R5, ≥ 8 pairs): INCOMPLETE (5 complete pairs < 8 : tests reported, no verdict)

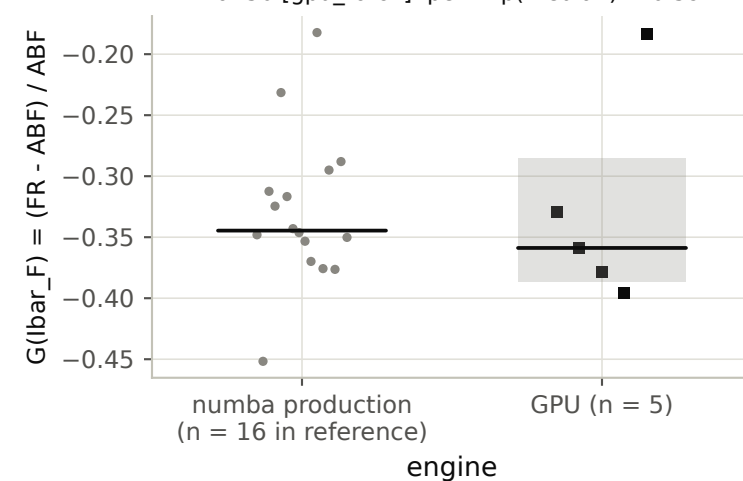
lta150 [gpu_torch]: perm. $p(\text{median}) = 0.56$, MWU $p = 0.55$



lta150 [gpu_torch]: perm. $p(\text{median}) = 0.35$, MWU $p = 0.45$



lta150 [gpu_torch]: perm. $p(\text{median}) = 0.59$



● numba production, one point per seed

■ torch backend, one point per seed

■ central 95 % of the backend median under H0 (permutation null)

— median